Question

A certain compound $\mathbf{X}$ contains only two elements, $\mathbf{A}$ and $\mathbf{B}$.

Under high pressure, $\mathbf{X}$ forms an atomic crystal. This crystal is orthorhombic, in which $\mathbf{A}$ is four-coordinated and $\mathbf{B}$ is two-coordinated, $a = 622\ \mathrm{pm}, \mathrm{b} = 435\ \mathrm{pm}, \mathrm{c} = 607\ \mathrm{pm}, \rho = 3.56\ \mathrm{g} \cdot \mathrm{cm}^{-3}, \mathrm{Z} = 8$.

$\mathbf{X}$ can form a cubic molecular crystal under low temperature and high pressure, $\mathrm{Z} = 4$, and the molecules cannot rotate freely.

Under high pressure, $\mathbf{X}$ can also form a tetragonal molecular crystal with cell parameters $a = 354\ \mathrm{pm}, \mathrm{c} = 413\ \mathrm{pm}, \mathrm{Z} = 2$. In a proper unit cell, if the $\mathbf{A}$ atoms are at the vertices of the unit cell, then the coordinates of two $\mathbf{B}$ atoms are $(0.266, 0.266, 0)$ and $(0.766, 0.234, 0.5)$, respectively.

Please select all the correct options from the following propositions:

1. $\mathbf{X}$ molecule has a center of symmetry

2. The lattice type of the cubic crystal of $\mathbf{X}$ is simple cubic

3. The tetragonal crystal of $\mathbf{X}$ has a center of symmetry

4. The tetragonal crystal of $\mathbf{X}$ has a twofold rotation axis passing through the $\mathbf{B}$ atoms

5. The tetragonal crystal of $\mathbf{X}$ does not have a twofold rotation axis not passing through the $\mathbf{A}$ atoms

A. 1,3,4

B. 1,3,5

C. 1,4,5

D. 2,3,4

E. 2,3,5

F. 2,4,5

G. 1,2,3,4

H. 1,2,3,5

I. 1,2,4,5

J. 1,3,4,5

K. 1,2,3,4,5

Answer

Correct Answer: G

Detailed Explanation

First, deduce the chemical formula of $\mathbf{X}$:

Unit cell volume $V = a \times b \times c = 622\ \mathrm{pm} \times 435\ \mathrm{pm} \times 607\ \mathrm{pm} = (1.643 \times 10^{8})\ \mathrm{pm}^3 \approx (1.643 \times 10^{-22})\ \mathrm{cm}^3$

Molar mass of $\mathbf{X}$ $M = \frac{\rho \times V \times N_A}{Z} = \frac{3.56 \times 1.643 \times 10^{-22} \times 6.022 \times 10^{23}}{8} \approx 44.0\ \mathrm{g \cdot mol^{-1}}$

CHECKPOINT 2 PTS

Molar mass of $\mathbf{X}$ $M \approx 44.0\ \mathrm{g \cdot mol^{-1}}$

Since the total number of chemical bonds is conserved, the coordination number of $\mathbf{A}$ to $\mathbf{B}$ is equal to the coordination number of $\mathbf{B}$ to $\mathbf{A}$, hence $4 \times n_A = 2 \times n_B$.

Therefore, the atomic ratio of $\mathbf{A}$ and $\mathbf{B}$ is 1:2

CHECKPOINT 1 PTS

The atomic ratio of $\mathbf{A}$ and $\mathbf{B}$ is 1:2

Compounds that meet the above conditions may be $\mathrm{CO_2}$ or $\mathrm{N_2O}$, but it is difficult for N_2O to achieve four-coordination of oxygen elements, so the chemical formula of $\mathbf{X}$ is $\mathrm{CO_2}$, $\mathbf{A}$ is C, and $\mathbf{B}$ is O

CHECKPOINT 1 PTS

$\mathbf{X}$ is $\mathrm{CO_2}$, $\mathbf{A}$ is C, $\mathbf{B}$ is O

The $\mathrm{CO_2}$ molecule has a center of symmetry, so proposition 1 is correct.

CHECKPOINT 1 PTS

The $\mathrm{CO_2}$ molecule has a center of symmetry, proposition 1 is correct

The molecular crystal of the cubic crystal system formed by $\mathrm{CO_2}$ has 4 $\mathrm{CO_2}$ molecules in its proper unit cell. If it is a face-centered cubic lattice, the spatial orientation of the 4 $\mathrm{CO_2}$ molecules must be consistent, which will lead to the loss of symmetry of the cubic crystal system. Therefore, under the conditions of the problem, the cubic crystal of $\mathbf{X}$ can only be a simple cubic lattice. Proposition 2 is correct.

CHECKPOINT 1 PTS

The cubic crystal of $\mathbf{X}$ is a simple cubic lattice, proposition 2 is correct

Next, analyze the tetragonal crystal of $\mathbf{X}$. The ungiven atomic coordinates can be obtained with the help of the symmetry of the tetragonal crystal system and the molecular symmetry of $\mathrm{CO_2}$.

Let $\mathbf{C1}(0, 0, 0)$, $\mathbf{O1}(0.266, 0.266, 0)$, $\mathbf{O2}(0.766, 0.234, 0.5)$

$\mathbf{C1}$ and $\mathbf{O1}$ belong to the same molecule, then we can get $\mathbf{O3}(0.734, 0.734, 0)$

According to the characteristic symmetry elements of the tetragonal crystal system, the crystal must have a four-fold rotation axis, inversion axis, or screw axis in the z direction. The $\mathrm{CO_2}$ molecule does not satisfy the requirements of the four-fold rotation axis or inversion axis in the z direction, so it can only be a 4_2 screw axis in the z direction. From this, we can determine that $\mathbf{C2}(0.5, 0.5, 0.5)$, $\mathbf{O4}(0.234, 0.766, 0.5)$

CHECKPOINT 2 PTS

$\mathbf{C2}(0.5, 0.5, 0.5)$, $\mathbf{O3}(0.734, 0.734, 0)$, $\mathbf{O4}(0.234, 0.766, 0.5)$

Then analyze the symmetry elements. The positions of all C atoms in the crystal are centers of symmetry, so proposition 3 is correct.

CHECKPOINT 1 PTS

The tetragonal crystal has a center of symmetry, proposition 3 is correct

The straight line where the $\mathrm{CO_2}$ molecule is located is a two-fold rotation axis, so proposition 4 is correct.

CHECKPOINT 1 PTS

Proposition 4 is correct

The straight line through $x = 0, y = 0.5$ or $x = 0.5, y = 0$, and parallel to the z axis is a two-fold rotation axis, which does not pass through $\mathbf{A}$, so proposition 5 is incorrect.

CHECKPOINT 1 PTS

Proposition 5 is incorrect

Based on the above analysis, the correct options are 1, 2, 3, 4